

# Field Operator Knowledge Record

Operator: Bob Patil | Senior Maintenance Tech (22 years)  
Date Recorded: 2026-03-14 | Area: Cooling Infrastructure

## Observation 1: Pump P-101 Vibration and Valve V-20 Root Cause

I have been working on this pump for fifteen years and I can tell you for certain. Every single time P-101 starts vibrating badly during hot weather or high thermal loads, it is because valve V-20 is not fully open. The previous shift tends to partially close it during line checks and forgets to re-open it. The fix is simple: go to valve V-20 handwheel and open it two full rotations counterclockwise. Within five minutes the vibration drops completely and the pump settles to normal. I have done this over forty times in fifteen years. Do not schedule a full teardown until you have checked V-20 first.

Asset: P-101 (PUMP-101) | Upstream valve: V-20 (VALVE-020)  
Issue: Vibration under high thermal load  
Fix: Open V-20 two full rotations counterclockwise  
Confidence: HIGH (40+ occurrences confirmed over 15 years)

## Observation 2: Boiler B-04 Has No Backup Feed Pump

People forget that B-04 has no backup feed pump. If P-101 goes down for any reason, you have eight minutes before the boiler overheats. In practice during summer above 38 degrees Celsius ambient, it is more like five to six minutes. When P-101 alarms, the boiler team must be notified immediately by radio. There is no automatic interlock. The interlock was deferred in the 2019 upgrade. Pending work order: WO-2019-0847.

Asset: B-04 (BOILER-04) | Feed pump: P-101  
Risk: No automatic P-101-trip to B-04-shutdown interlock  
Action: Manual radio call required within 5 to 8 minutes of P-101 trip  
Pending: WO-2019-0847

## Observation 3: Valve V-20 Stiff Handwheel Since 2024

The handwheel on V-20 has been stiff since the 2024 annual shutdown. They replaced the packing but did not properly torque the gland follower. You need two hands to turn it. Use the extended handle T-EXT-044 from the C-block tool cabinet, second shelf. Without it you need two operators to turn V-20 in an emergency, which is a safety issue.

Asset: V-20 (VALVE-020)  
Issue: Stiff handwheel due to improperly torqued gland follower after 2024 packing job  
Workaround: Extended handle T-EXT-044, C-block tool cabinet, shelf 2

## Observation 4: B-04 Emissions Spike Above 80 Percent Load

When B-04 runs above 80 percent capacity during winter peak demand, CO2 emissions climb well above the baseline 320 ppm. I have seen 410 ppm on cold days when all three boilers run together. The sensor has a 6-second lag so real peak is probably higher. Someone should verify the permit limit from the 2025 environmental compliance review.

Asset: B-04 (BOILER-04)  
Baseline: 320 ppm CO2 | Observed peak: ~410 ppm at above 80% load  
Sensor lag: 6 seconds. Real peak likely higher.